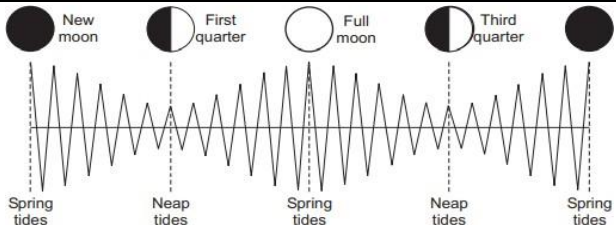
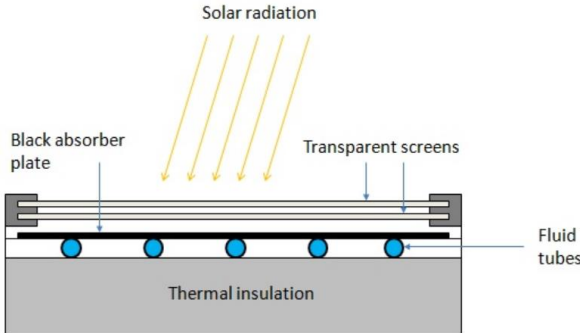


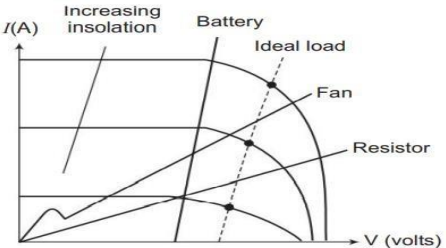
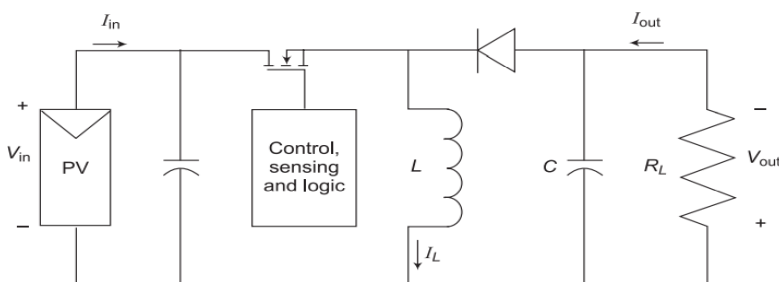
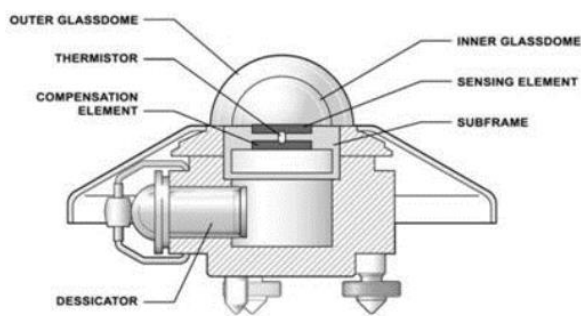
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY												
SIXTH SEMESTER B.TECH DEGREE EXAMINATION, JUNE 2022												
Course Code: EET322												
Course Name: RENEWABLE ENERGY SYSTEMS												
Max. Marks: 100		Duration: 3 Hours										
PART A												
		Answer all questions, each carries 3 marks.		Marks								
1		The Kyoto Protocol is an international agreement linked to the United Nations Framework Convention on Climate Change, which commits its Parties by setting internationally binding emission reduction targets. The Protocol places a heavier burden on developed nations – Australia, USA, European Countries, under the principle of "common but differentiated responsibilities", recognizing that developed countries are principally responsible for the current high levels of GHG emissions in the atmosphere as a result of more than 150 years of industrial activity.		(3)								
2		Rising Seas Changes in rainfall patterns Increased likelihood of extreme events Melting of the ice Melting glaciers		(3)								
3		<table><tr><th>Flat Plate</th><th>Concentrating Type</th></tr><tr><td>Absorbs both beam and diffused radiation</td><td>Mainly uses beam radiation</td></tr><tr><td>Simple in construction & does not require sun tracking</td><td>Complicated construction due to solar tracking</td></tr><tr><td>Absence of optical concentration , hence the area from which heat is lost is large</td><td>High temperatures can be attained due to concentration of radiation.</td></tr></table>	Flat Plate	Concentrating Type	Absorbs both beam and diffused radiation	Mainly uses beam radiation	Simple in construction & does not require sun tracking	Complicated construction due to solar tracking	Absence of optical concentration , hence the area from which heat is lost is large	High temperatures can be attained due to concentration of radiation.	(3)	
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4		(a) Cross section (b) Single crystal cell (off square) (c) Multicrystalline cell (square)		(3)								

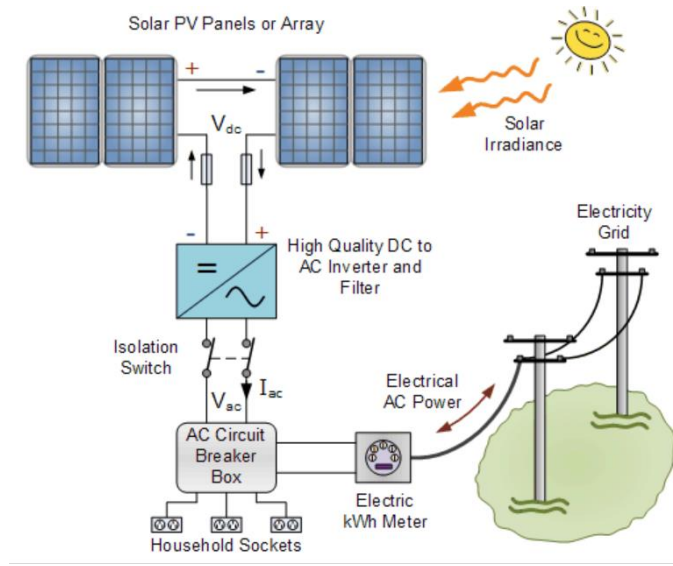
5	Windy Conditions Site Topography Geological Conditions Industrial Area	(3)
6	A German physicist Albert Betz concluded in 1919 that no wind turbine can convert more than 16/27 (59.3%) of the kinetic energy of the wind into mechanical energy turning a rotor. To this way, this is known as the Betz Limit or Betz'Law. The theoretical maximum power efficiency of any design of wind turbine is 0.59 (i.e. no more than 59% of the energy carried by the wind can be extracted by a wind turbine). This is called the "power coefficient" and is defined as: $C_{pmax}=0.59$	(3)
7	 The sun's effect is similar but smaller in magnitude (about 2.2 times less than lunar), and with a period of 12 hours. When the sun, earth and moon are aligned (approximately) in conjunction, the lunar and solar tides are in phase, producing net tides of maximum range. These are the spring tides occurring twice per lunar month at times of both full and new moon. When sun-earth and moon-earth directions are perpendicular (in quadrature), the solar and lunar tides are out of phase producing net tides of minimum range. These are the neap tides that again occur twice per month at times of half-moon (first and third quarter cycle of the moon)	(3)
8	The deposition and growth of microorganisms inside the pipes of evaporator and condenser heat exchanger is known as biofouling . Biofouling reduces the heat transfer efficiency and thereby lowers the performance. Such biofouling is one of the major problems in OTEC design, since increasing the surface area available for heat transfer also increases the opportunity for microorganisms to attach themselves. Among the methods being tried to keep the fouling under control are mechanical cleaning by continuous circulation of close-fitting balls and the chemical cleaning (chlorination) by additives to the water.	(3)
9	Hydrogen storage in metal hydrides is possible because some metals and metal alloys are capable of storing gaseous hydrogen. In this process, the H atoms – i.e. hydrogen in dissolved form – are deposited in so-called "interstitial sites. Metal and gas form a compound, the result is a metal hydride.	(3)
10	RDF stands for Refuse Derived Fuel. This fuel is produced from combustible components that the industry calls Municipal Solid Waste – MSW for short. This waste, usually taken from industrial or commercial sites, is shred, dried, baled and then finally burned to produce electricity.	(3)
PART B		

Answer one full question from each module, each carries 14 marks.

Module I																										
11	a)	Quality of life Equality Diversity Social Cohesion Democracy & Governance																								
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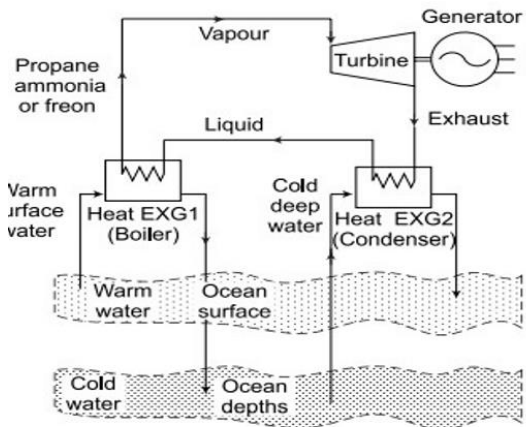
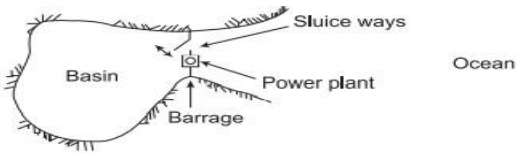
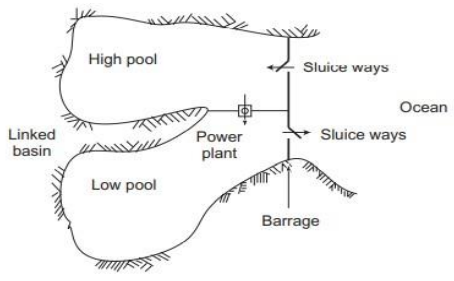
12	a)	• Advantages • i. Available in nature • ii. Free of cost • iii. Inexhaustible • iv. Little pollution • v. Low gestation period • Disadvantages • i. High cost of harnessing energy • ii. Uncertainty of availability • iii. Difficulty in transporting	(6)
	b)	• Burning of fossil fuels and fibers. • Cutting of forests. • Erruption of volcanoes. Exposure to high levels of air pollution can cause a variety of adverse health outcomes. It increases the risk of respiratory infections, heart disease and lung cancer. Both short and long term exposure to air pollutants have been associated with health impacts. More severe impacts affect people who are already ill	(4)
	c)	Sulfur oxides(SOx) : It is produced by volcanoes and in various industrial processes.Since coal and petroleum often contain sulfur compounds, their combustion generates sulfur dioxide. Further oxidation of SO₂,usually in the presence of a catalyst such as NO₂, forms H₂SO₄,and thus acid rain.These causes environmental impact. Nitrogen oxides(NOx): It is emitted from high temperature combustion. This reddish - brown toxic gas has a characteristic sharp, biting odor	(4)
		Module II	
13	a)	A flat plate collector normally comprises the following components: a plate to capture the solar energy by the glazing frame, flow tubes that permit the water to flow into it and gain the heat that is captured by the absorber plate, heat transfer losses, and a box enclosure can be controlled by adding insulation 	(6)
	b)	When a solar PV system is deployed for practical applications, the I-V characteristic keeps on changing with insolation and temperature. In order to receive maximum power the load must adjust itself accordingly to track the maximum power point. The I-V characteristics of PV system, along with some common loads, are shown in Fig. An ideal load is one that tracks the maximum power point.	(8)

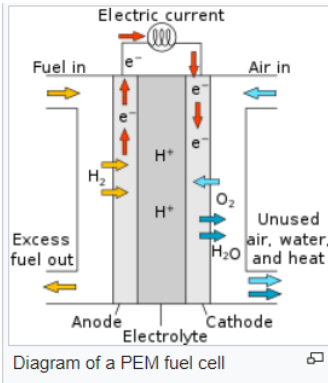
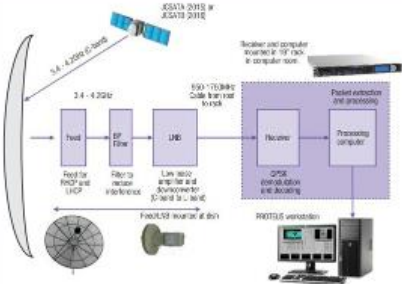
		  $V_{out} = \frac{D}{1-D} V_{in}$ where D is the duty cycle of the MOSFET, expressed as fraction ($0 < D < 1$). Details of operation and design of the converter may be found in any standard book of power electronics.	
		OR	
14	a)	Based on the Seebeck- or thermoelectric effect, a pyranometer is operated based on the measurement of a temperature difference between a clear surface and a dark surface. The black coating on the thermopile sensor absorbs solar radiation, while the clear surface reflects it. Hence, less heat is absorbed on the clear surface. The potential difference created in the thermopile owing to the temperature gradient between the two surfaces reveals information about the amount of solar radiation. The voltage produced by the thermopile can also be measured using a potentiometer. While most pyranometers generate an analog signal, some come with a smart digital interface. 	(8)

b)	A grid-interactive solar system generates clean power from the unlimited energy of sunlight. Grid-interactive inverters can also use other renewable sources such as wind or hydro power. When the sun shines, the PV system charges its deep cycle solar batteries and feeds all excess clean power into the electricity grid via the inverter. When there is no solar power because it's night, the system can draw power from the electricity grid. When there is a power failure, the inverter draws clean power from the solar panels and batteries to meet the electricity needs of your home or business. Then it is a stand-alone system. 	(6)																																			
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		power system. A general block diagram of a grid connected WECS is shown in Fig	
		OR	
16	a)	Lift & Drag Forces The <u>Lift Force</u> is perpendicular to the direction of motion. We want to make this force BIG. The <u>Drag Force</u> is parallel to the direction of motion. We want to make this force small. © 1998 www.WINDPOWER.de $\alpha = \text{low}$ $\alpha = \text{medium} < 10 \text{ degrees}$ $\alpha = \text{High Stall!!!}$	(8)
	b)		(6)

		Module IV																
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	b)	In a hybrid cycle combines the features of both the closed-cycle and open-cycle systems. In a hybrid OTEC system, warm seawater enters a vacuum chamber where it is flash- evaporated into steam, similar to the open- cycle process. The steam vapourises the ammonia working fluid of a closed-cycle loop on the other side of an ammonia vapourizer. The vapourised fluid then drives a turbine coupled to an electric generator producing electricity. Hybrid cycle OTEC plant																
		OR																
18	a)	In closed cycle (also known as Anderson cycle) plant, warm surface water is used to evaporate a low boiling point working fluid such as ammonia, freon or propane. The vapor flows through the turbine and is then cooled and condensed by cold water pumped from the ocean depths. Because of low quality heat a large surface areas of heat exchangers (evaporator and condenser) are required to transfer significant amount of heat and large amount of water need to be circulated. The operating pressures of the working fluid at the boiler/evaporator and condenser are much higher and its specific volume is much lower as compared to water in open cycle system. Such pressures and specific volumes result in turbine that is much smaller in size and hence less costly as compared to that in open cycle system.		(6)														

			
	b)	4 classifications: i) Single-basin single-effect plant (ii) Single-basin double-effect plant (iii) Double-basin with linked-basin operation (iv) Double-basin with paired-basin operation  (a) Layout of single basin tidal energy conversion scheme 	(8)
		Module V	
19	a)	In the process, there are several factors that affect the biochemical as well as physiochemical process. The most important factors are temperature, pH, OLR, C:N ratio, etc., that has to be controlled to be the adequate level for the efficient anaerobic digestion process and the methane yield	(6)

b)	 Diagram of a PEM fuel cell Proton-exchange membrane fuel cells (PEMFC), also known as polymer electrolyte membrane (PEM) fuel cells, are a type of fuel cell being developed mainly for transport applications, as well as for stationary fuel-cell applications and portable fuel-cell applications. Their distinguishing features include lower temperature/pressure ranges (50 to 100 °C) and a special proton-conducting polymer electrolyte membrane. PEMFCs generate electricity and operate on the opposite principle to PEM electrolysis, which consumes electricity. They are a leading candidate to replace the aging alkaline fuel-cell technology, which was used in the Space Shuttle	(8)
	OR	
20 a)	Spacecraft that orbit Earth, called satellites, are close enough to the Sun that they can often use solar power. These spacecraft have solar panels which convert the Sun's energy into electricity that powers the spacecraft. Typical research satellites operate with only 200 to 800 watts of electricity generated by sunlight and 'solar cells'. Solar cells can be attached directly to the outer surface of a satellite, or can be found on 'solar panels' that the satellite deploys after it reaches its orbit. 	(6)
b)	This is the first fixed-dome biogas plant was introduced in the form of the Janta Model Biogas Plant by Gobar Gas Research Station, Ajitwal in 1978. The main feature of this model is that the digester and the gas holder are integrated parts of brick masonry structure. The digester is made of a shallow well having a dome-shaped roof on it. The inlet and outlet tanks are connected with the digester through large chutes which are called displacement chambers. The gas pipe is fitted on the crown of the masonry dome and there is an opening on the outlet	(8)

wall of the outlet displacement chamber for the discharge of spent digested slurry. The size of these plants is limited to 15 m³ par day. The different components of Janta Model Biogas Plant are shown

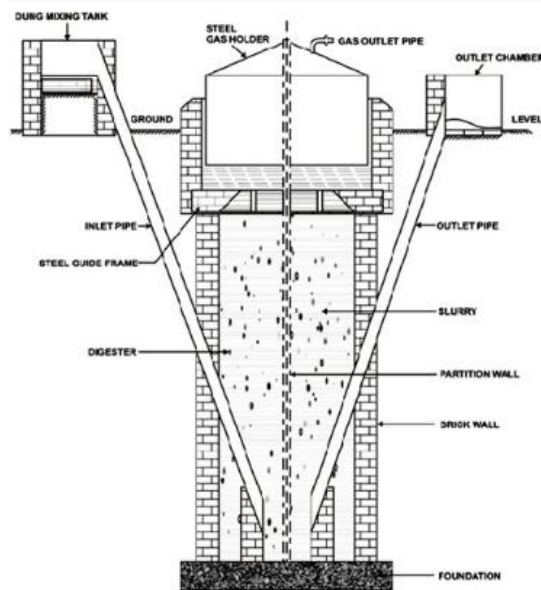


Fig. 6.1 KVIC model biogas plant

